

# **User Manual**

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## Introduction

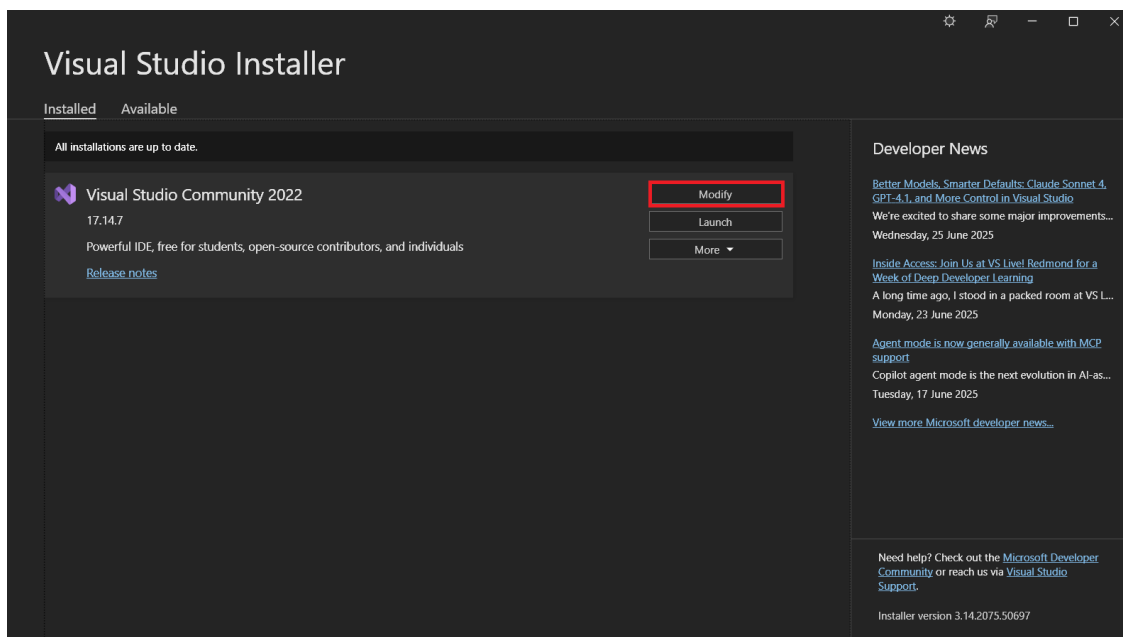
After getting all the necessary files and library, this guide will show you how to use the app from installing the required **C++ components** in **Visual Studio** to performing basic mathematical operations.

## Installing C++ Components in Visual Studio

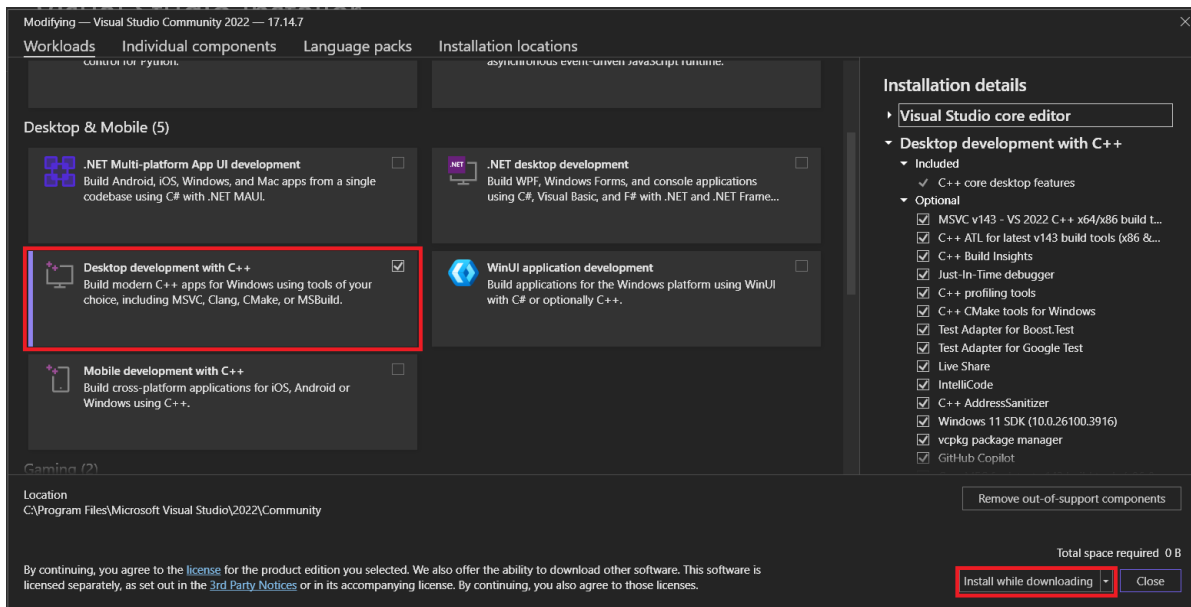
The software of choice here is **Visual Studio**.

### Steps:

1. Open **Visual Studio installer**
2. Click **Modify**



- Click **Desktop development with C++** as this will install the necessary components and click **Install while downloading**

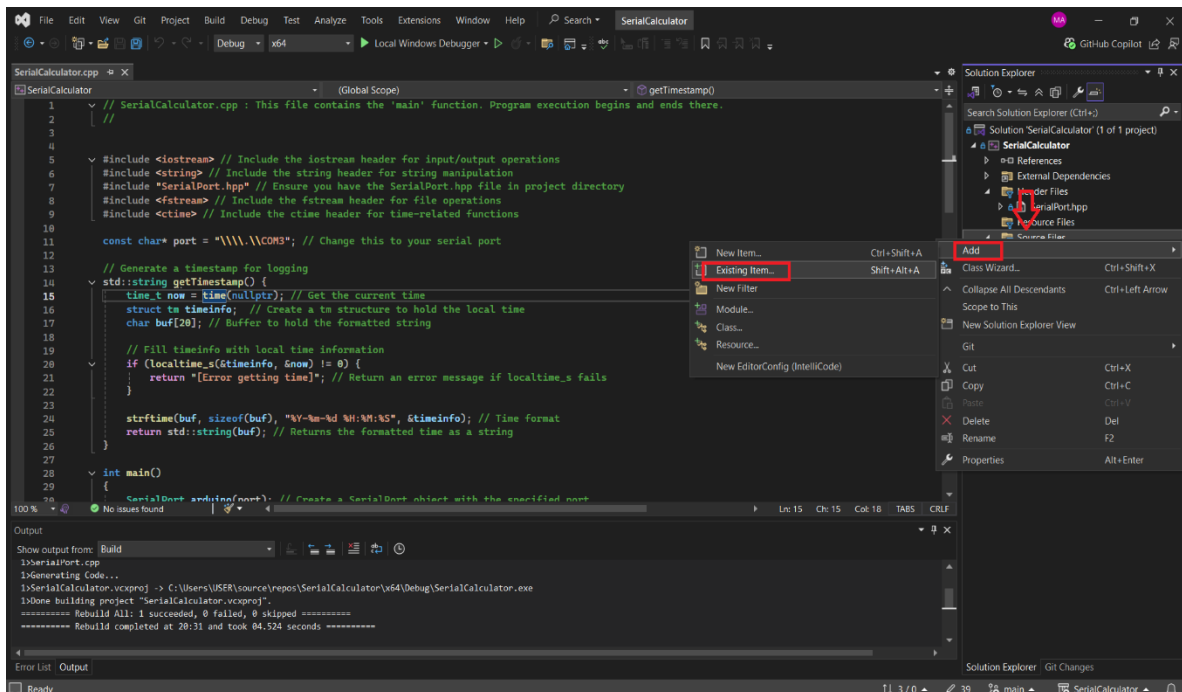


## Importing SerialPort Library (Console or Visual Studio)

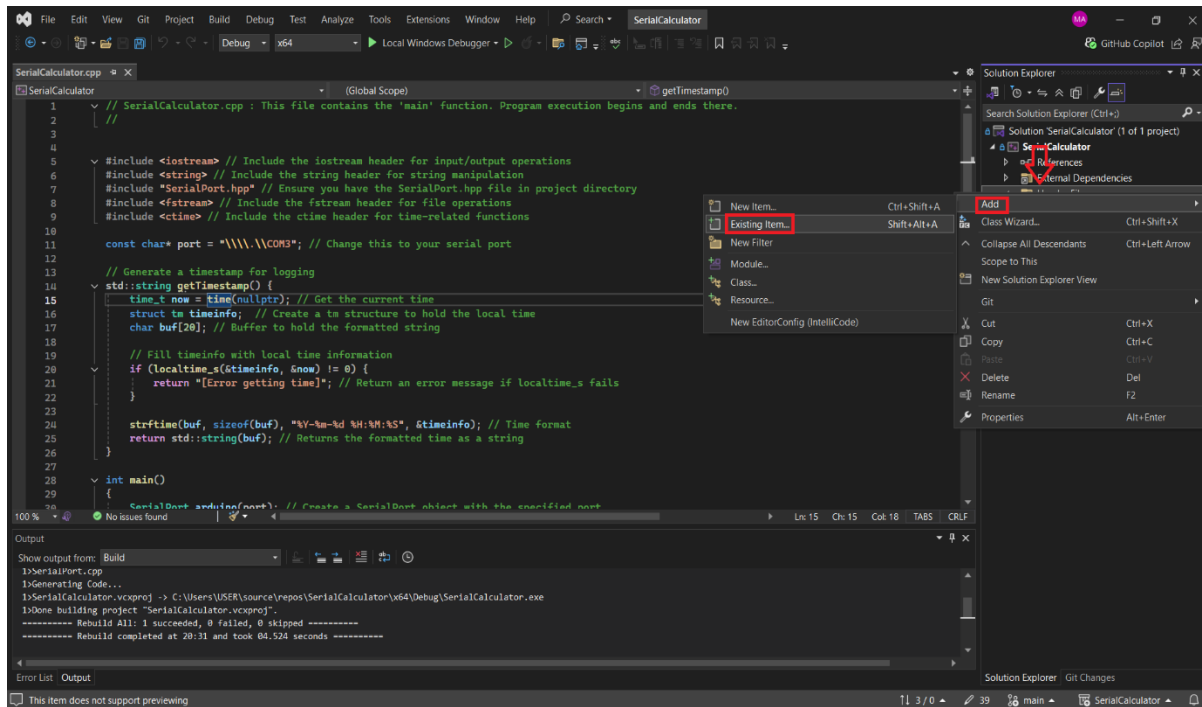
This can be done in two ways:

### 1. Import in Visual Studio

- Importing **SerialPort.cpp**



- Importing header file: **SerialPort.hpp**

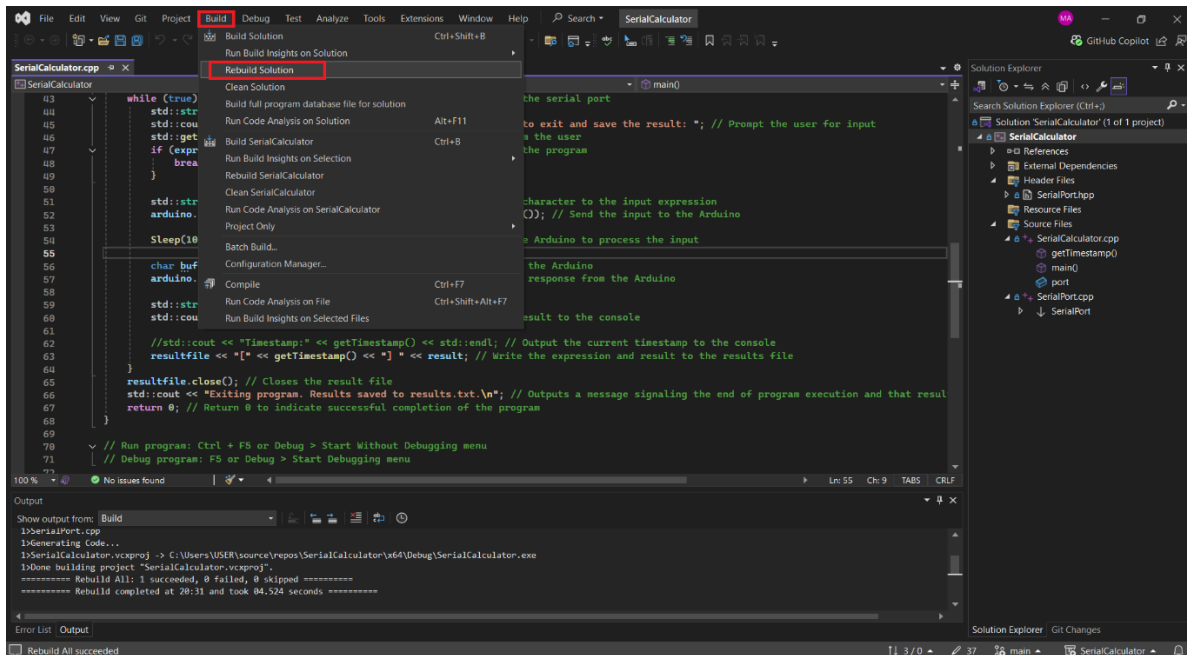


2. Extracting the **SerialPort.zip**, copy and paste the **SerialPort.cpp** and **SerialPort.hpp** directly into the **project folder**

## Building the Calculator

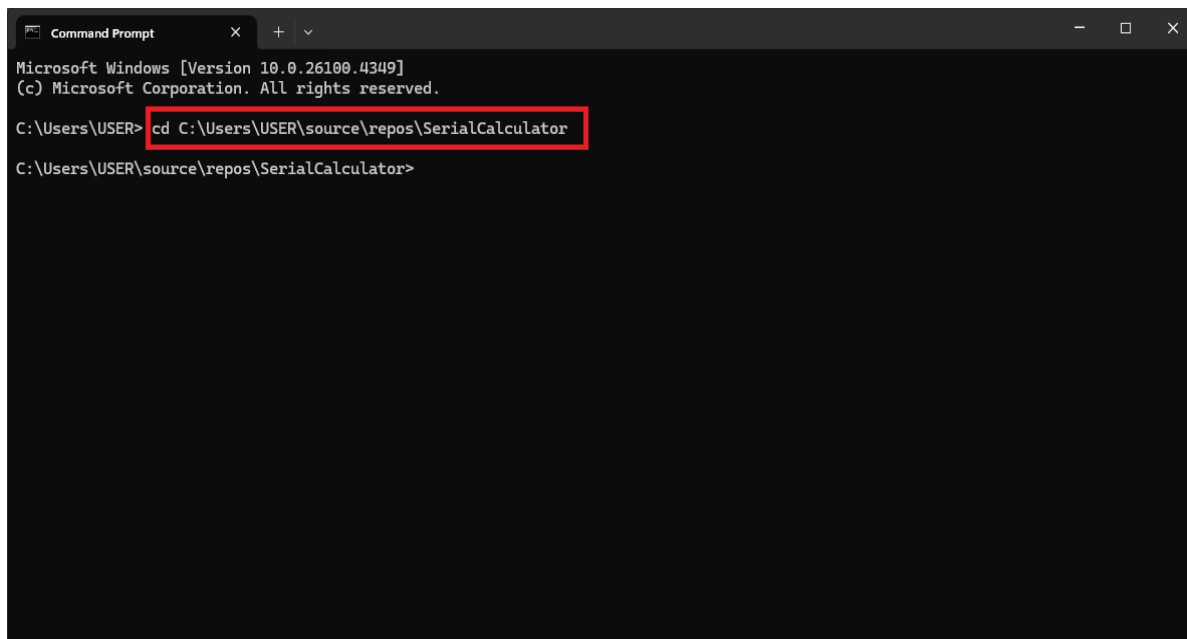
Two methods can be used to do this, either through Visual Studio or the Console App:

1. Visual Studio

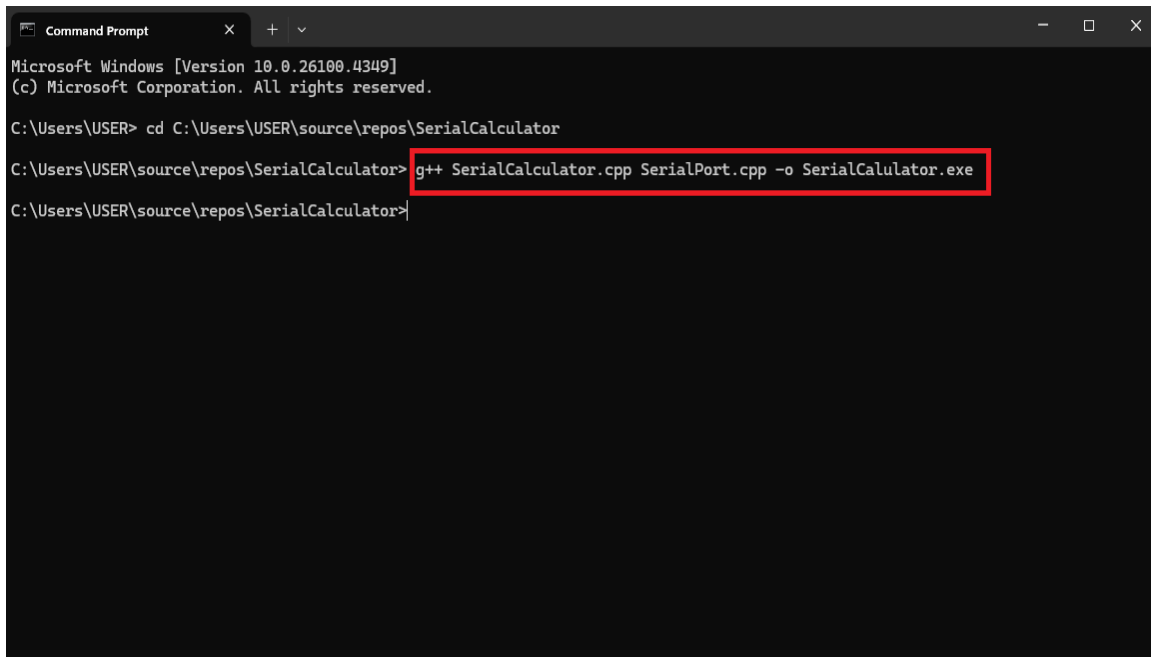


## 2. Console App

- Change **directory** to the **project directory**



- Creating **executable(.exe)** file



```
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

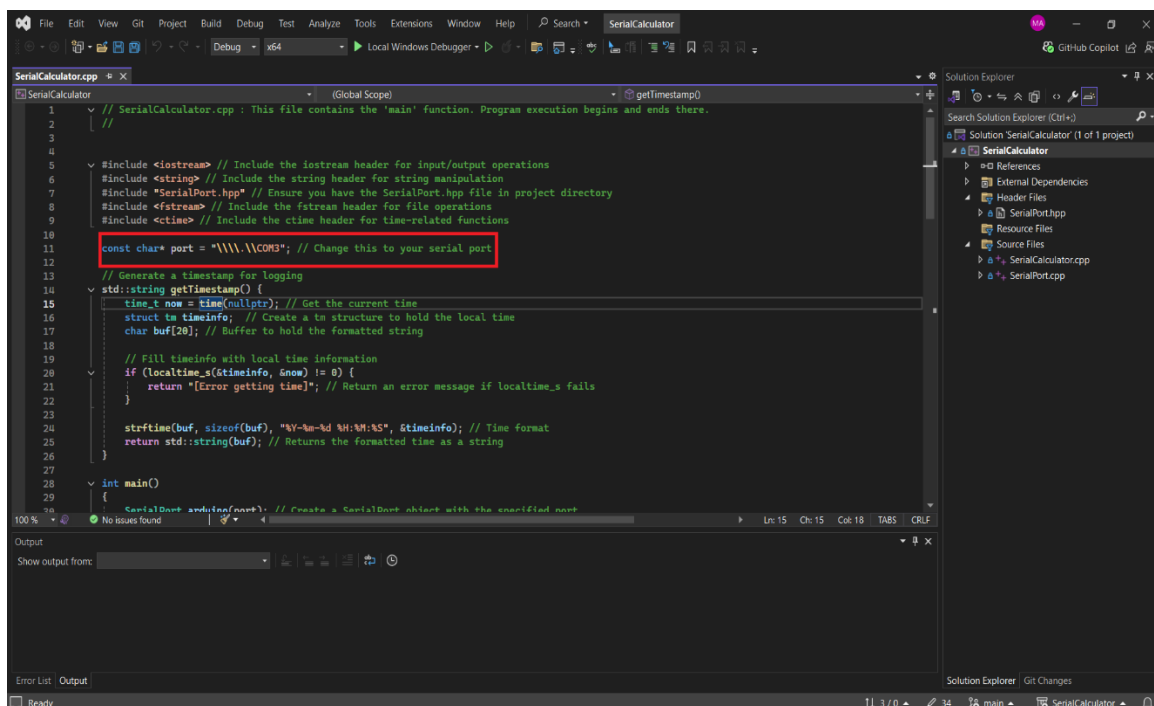
C:\Users\USER> cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator> g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator>
```

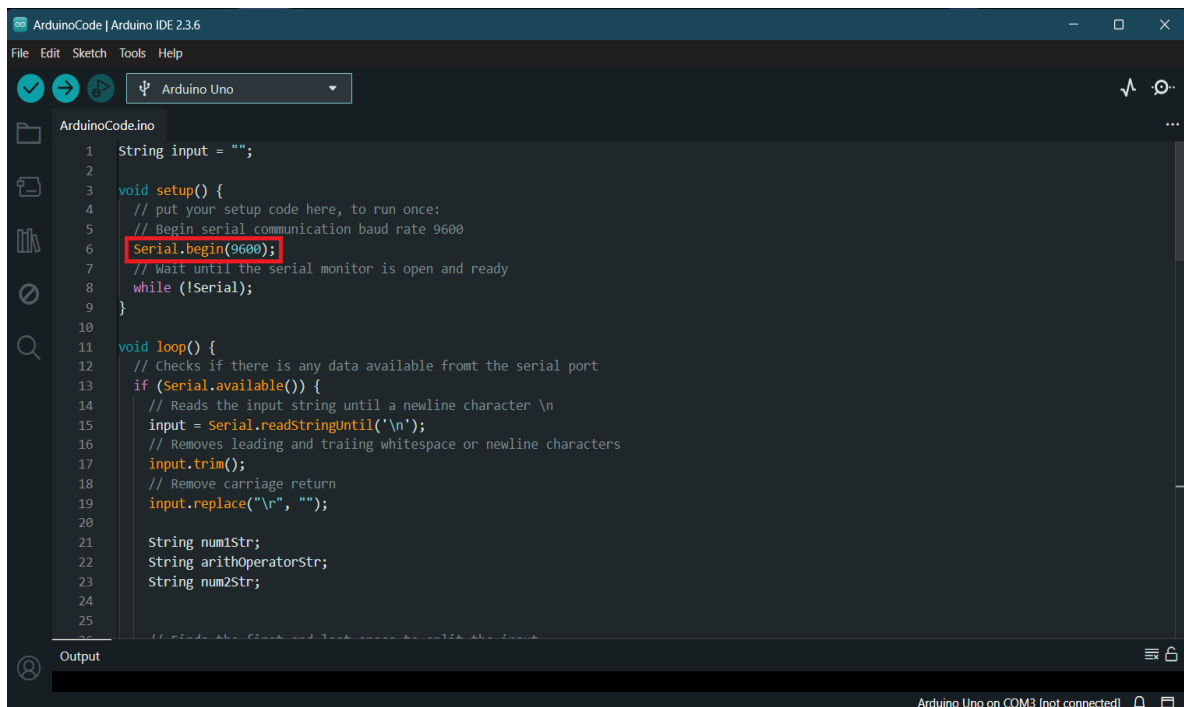
## Setting Up the Connection — COM port, baud rate

1. Selecting the COM port — Type the COM port where the microcontroller is connected (Open Arduino IDE to check if unsure)

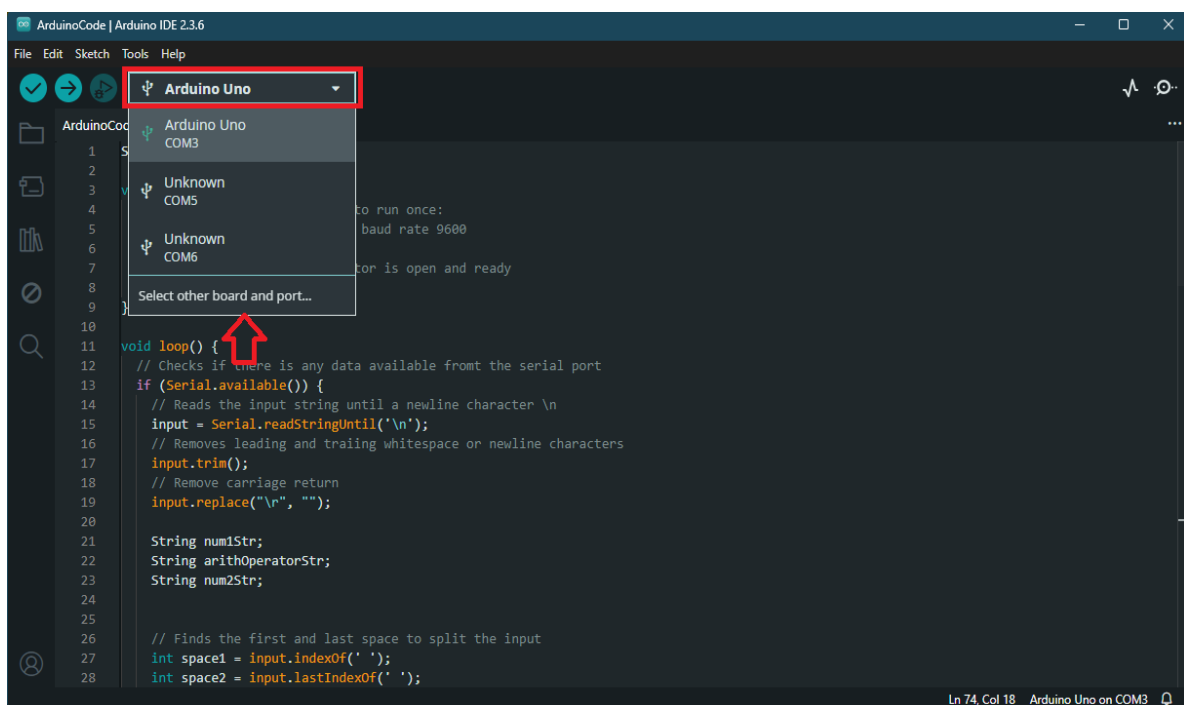


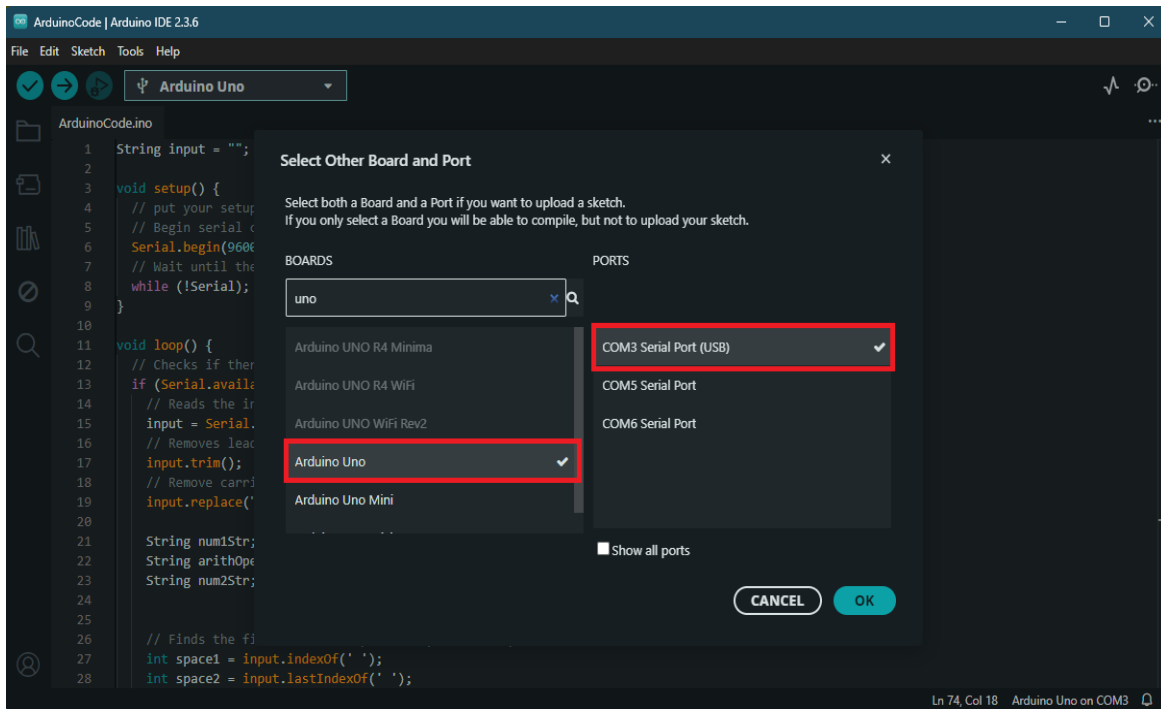
```
1 // SerialCalculator.cpp : This file contains the 'main' function. Program execution begins and ends there.
2 //
3
4
5 #include <iostream> // Include the iostream header for input/output operations
6 #include <string> // Include the string header for string manipulation
7 #include "SerialPort.hpp" // Ensure you have the SerialPort.hpp file in project directory
8 #include <fstream> // Include the fstream header for file operations
9 #include <ctime> // Include the ctime header for time-related functions
10
11 const char* port = "\\\\.\\COM3"; // Change this to your serial port
12
13 // Generate a timestamp for logging
14 std::string getTimestamp() {
15     time_t now = time(nullptr); // Get the current time
16     struct tm timeinfo; // Create a tm structure to hold the local time
17     char buf[20]; // Buffer to hold the formatted string
18
19     // Fill timeinfo with local time information
20     if (localtime_s(&timeinfo, &now) != 0) {
21         return "[Error getting time]"; // Return an error message if localtime_s fails
22     }
23
24     strftime(buf, sizeof(buf), "%Y-%m-%d %H:%M:%S", &timeinfo); // Time format
25     return std::string(buf); // Returns the formatted time as a string
26 }
27
28 int main()
29 {
30     SerialDont arduino(conn); // Create a SerialDont object with the specified port
31 }
```

2. Entering **baud rate**

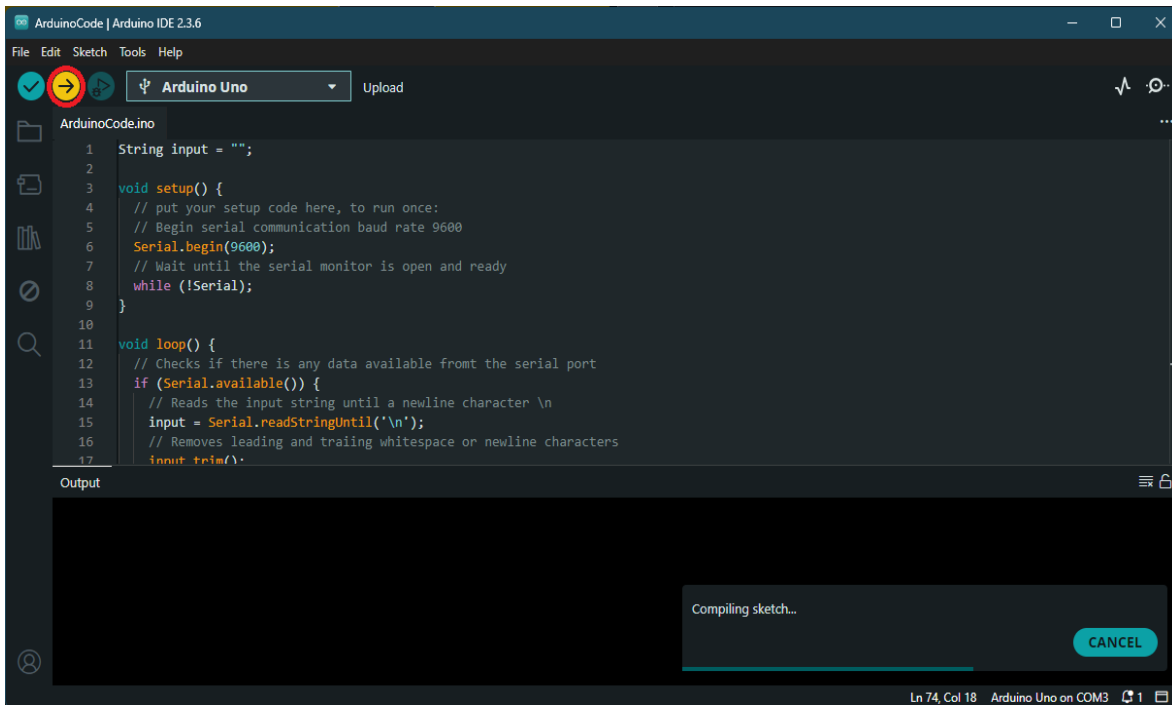


## Selecting the Arduino Board to the Microcontroller





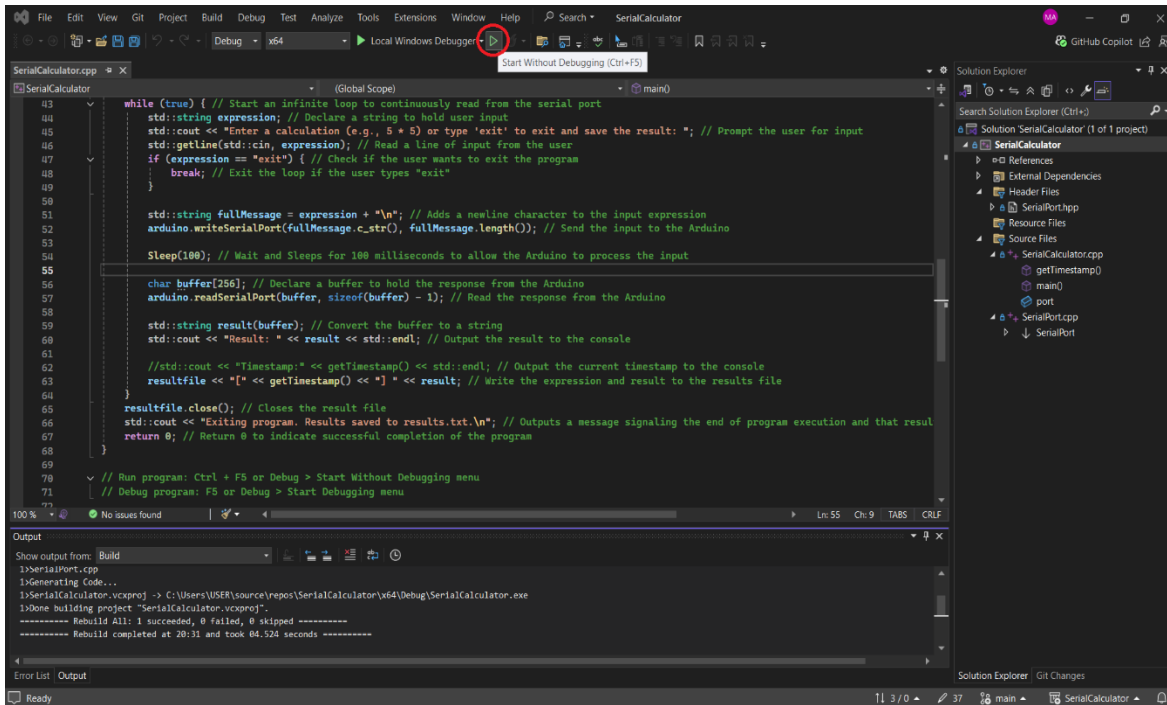
## Uploading *ArduinoCode.ino* to Microcontroller



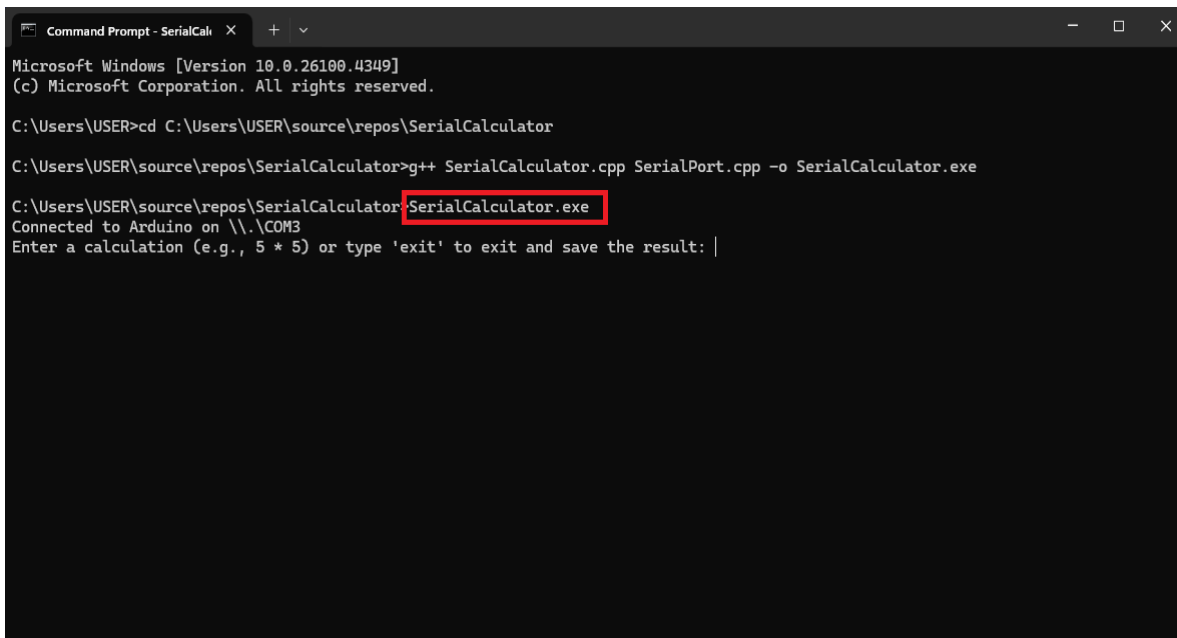
## Running the App — via Visual Studio | Console | .exe file

The app can be run through three different ways:

## 1. Visual Studio



## 2. Console app



## 3. Project folder



## Entering a Mathematical Expression

Typically, there are two ways for the expression to be entered that would be considered valid by the microcontroller and it is of this format: `<number1> <operator> <number2>`

### 1. Sample entry 1

```
Command Prompt - SerialCalv  x  +  v
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
Connected to Arduino on \\.\COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9 * 5
Result: Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

### 2. Sample entry 2

```
Command Prompt - SerialCal X + -
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
Connected to Arduino on \\.\COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9 * 5
Result: Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 9*5
Result: Result: 9 * 5 = 45

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: |
```

## Saving the Result

Once the user has completed the desired operations, to save the **results** from the calculations, the user must type the command **'exit'** to indicate to the program that the set of operation is complete. It then saves the operations to a **results.txt** files in **the project folder**.

**Note:** Abruptly **closing** the **command window/Console App** would result in **failure** of the program in **saving** the previous operations to the **file**.

All **previous operations** are shown in the file.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.4349]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd C:\Users\USER\source\repos\SerialCalculator

C:\Users\USER\source\repos\SerialCalculator>g++ SerialCalculator.cpp SerialPort.cpp -o SerialCalculator.exe

C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
ERROR!!!
Failed to connect to Arduino on\\.\COM3

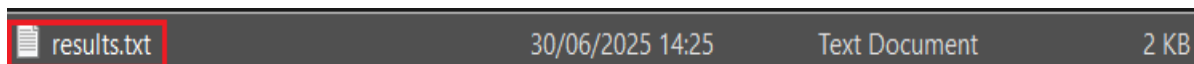
C:\Users\USER\source\repos\SerialCalculator>SerialCalculator.exe
Connected to Arduino on \\.\COM3
Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: 6*8
Result: Result: 6 * 8 = 48

Enter a calculation (e.g., 5 * 5) or type 'exit' to exit and save the result: exit
Exiting program. Results saved to results.txt.

C:\Users\USER\source\repos\SerialCalculator>
```

## Checking the *result.txt* File Location

To check the **history** of operations previously performed, go to the **project folder** and find the **results.txt** file. You can open the file with any compatible software including the native **notebook** software that comes pre-installed with **Windows**.



## Error Messages

- **“Error: Division by zero”** – Division by zero attempted
- **“Error: Invalid operator”** – Unsupported operator used
- **“Error: No valid operator found”** – No operator detected in input
- **“Error: Invalid input format”** – Incorrect input string

## Troubleshooting

### Common Issues

**“Failed to connect to Arduino”**

- **Cause:** Incorrect **COM** port, Arduino not connected or Serial monitor still open in Arduino IDE
- **Solution:**
  - Close the serial monitor in Arduino IDE
  - Ensure corrected port is selected by updating port variable in code (this can be verified via the Arduino IDE)
  - Ensure Arduino is connected
  - Ensure the proper drivers are installed, if not, check Device Manager

### “Could not open results file”

- **Cause:** File permissions or disk space issues
- **Solution:**
  - Check available disk space
  - Ensure directory is writable
  - Run application as administrator

### Arduino not responding

- **Cause:** Arduino not programmed or serial connection issues
- **Solution:**
  - Verify Arduino firmware is uploaded
  - Check serial monitor in Arduino IDE
  - Try different USB cable or port

### Incorrect Calculation

- **Cause:** Input parsing issues or precision limitations
- **Solution:**
  - Verify input format matches supported patterns (Format: `<number1> <operator> <number2>`)
  - Check for extra spaces or characters
  - Understand floating-point precision limits